

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

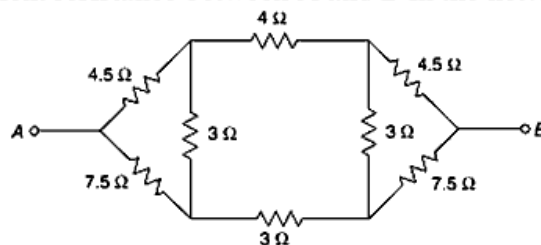
Q. 1 Solve Any Two of the following.

12

- A) Find an equivalent resistance between A and B in the network of Figure.

Understanding

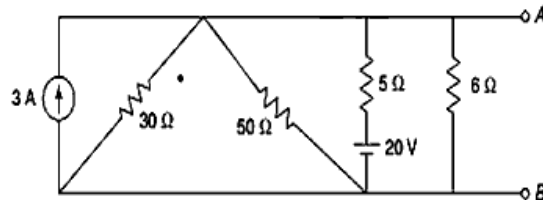
6



- B) Replace the circuit between A and B in Figure with a voltage source in series with a single resistor.

Remembering

6



- C) State and Explain classification of Elements with their examples.

Understanding

6

Q.2 Solve Any Two of the following.

12

- A) State and Explain Maximum Power Transfer Theorem.

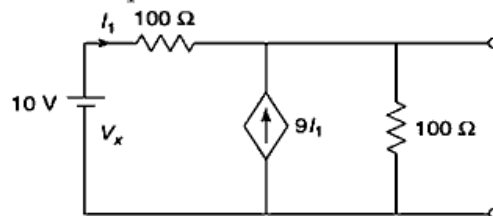
Remembering

6

- B) Determine Thevenin's equivalent network.

Understanding

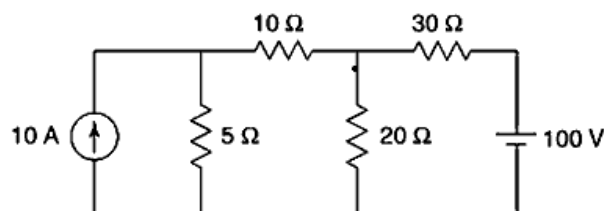
6



- C) Find the current through the 10Ω resistor by using superposition theorem.

Remembering

6



Q.3 Solve Any Two of the following.

12

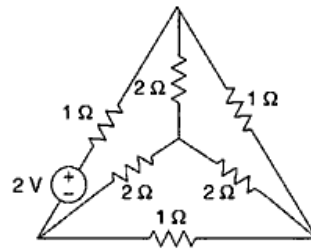
- A) Obtain transient response of driven R-C series circuit.
 B) Write down the tieset matrix and incidence matrix.

Remembering

6

Remembering

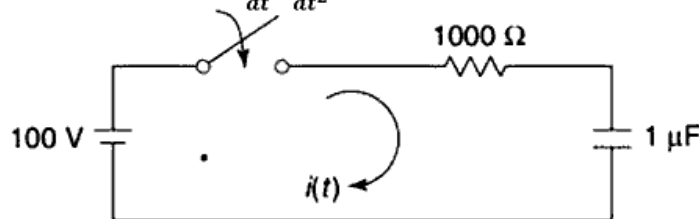
6



- C) In the network of Figure, the switch is closed at $t = 0$. With the capacitor uncharged, find value for i , $\frac{di}{dt}$, $\frac{d^2i}{dt^2}$ at $t = 0+$.

Understanding

6



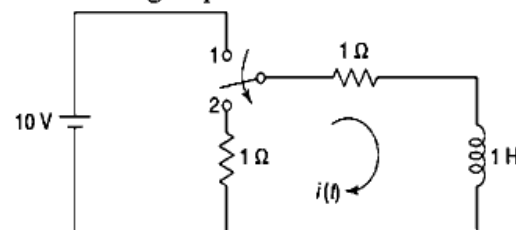
Q.4 Solve Any Two of the following.

12

- A) In the network of Figure, the switch is moved from the position 1 to 2 at $t = 0$, steady-state condition having been established in the position 1. Determine $i(t)$ for $t > 0$ using Laplace Transform.

Understanding

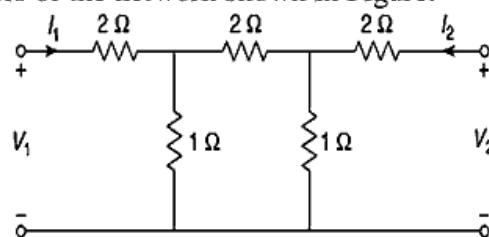
6



- B) Find Y-parameters of the network shown in Figure.

Understanding

6



- C) What is the physical significance of Pole and Zero in a Transfer Function?

Remembering

6

Q.5 Solve Any Two of the following.

12

- A) Explain High pass filter and band pass filter.
 B) What is meant by resonance in series RLC circuit? Derive equation for resonant frequency.
 C) A series RLC circuit has the following parameter values: $R = 10 \Omega$, $L = 0.01 \text{ H}$, $C = 100 \text{ mF}$. Compute the resonant frequency, bandwidth, and lower and upper frequencies of the band width.

Understanding

6

Remembering

6

Understanding

6

*** End ***